W-B-820A
September 18, 1978
SUPERSEDING
W-B-820
January 10, 1966

FEDERAL SPECIFICATION

BELL, ELECTRICAL, LOW VOLTAGE; BUZZER ELECTRICAL, LOW VOLTAGE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a vibrating signal type, electrical bell and a vibrating signal type, electrical buzzer for use with low voltage alternating current (ac) or direct current (dc) in interior communication systems.

1.2 Classification. Bells and buzzers covered by this specification shall be of the following types and styles, as specified (see 6.2):

1.2.1 Types and styles.

Type I - Bell, electrical, low voltage.
Type II - Buzzer, electrical, low voltage.

Style A - dc.
Style B - ac.
Style C - ac and dc.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.

FSC 6350

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PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

Federal Standard:

Fed. Std. No. H28 - Screw-Thread Standards for Federal Services.
Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadephia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Forth Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-116 - Preservation-Packaging, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The bell and buzzer are used as signaling devices in interior communication systems and shall consist of a housing, an operating mechanism, and a sound mechanism. The bell and buzzer shall be suitable for surface mounting on wood and grounded or ungrounded metal.

3.2 First article. When specified (see 6.2), the contractor shall furnish one bell or buzzer of each type and style included under a specific contract for first article inspection and approval (see 4.3 and 6.5).

3.3 Standard product. Except where modified herein, the bells and buzzers shall be the regular commercial product of the manufacturer or his suppliers, all materials, components, equipment, and accessories shall be new and unused. All nuts, bolts, and screws shall have standard screw threads in accordance with Fed. Std. No. H28.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as

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opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specifically specified.

3.5 Electrical connections. Electrical connections shall be mechanically and electrically secured by either soldering, riveting, or crimping. The bell and buzzer shall be provided with either exposed or concealed stud or screw type terminals for connection to the low voltage power source.

3.6 Electrical rating. Unless otherwise specified (see 6.2), style A shall be rated for operation on 6 to 8 volts (V) dc; style B on 8 to 10 V, 60 Hertz (Hz) ac; and style C on 6 to 8 V 60 Hz ac and 3 to 6 V dc.

3.7 Type I, bell sound level. The sound level of the signal output from the bell at 10 feet shall be not less than 65 decibels (dB), when tested in accordance with 4.4. Means for adjustment of the tone or volume of the signal output shall not be required.

3.8 Type II, buzzer sound level. The sound level of the signal output from the buzzer at 3 feet shall be not less than 80 dB when tested in accordance with 4.4. Means shall be provided for adjustment of the tone and volume of the signal output.

3.9 Housing. The housing for the bell and buzzer shall be fabricated from sheet metal and shall consist of a base and a removeable cover, or the buzzer may be enclosed in its own inaccessable housing. The cover shall enclose the electrical mechanism.

3.10 Operating mechanism. The operating mechanism of the bell and the buzzer shall consist of an electromagnet, a steel vibrating armature, silver or silver alloy contacts, where necessary, and suitable springs and wiring connections.

3.11 Sounding mechanism. The bell shall be provided with an exposed gong having a diameter of approximately 2-1/2 inches which shall be fabricated from steel or brass alloys. The steel vibrating armature of the bell shall be constructed in the form of a clapper with an attached steel hammer for striking the gong to produce the bell sound. The buzzer shall produce the buzzer sound by the steel vibrating armature striking a solid member of the structure.

3.12 Dimensions. The overall dimensions of the bell shall be approximately 7 by 2-1/2 by 1-1/2 inches. The overall dimensions of the buzzer, with a minimum sound level of 80 dB, shall be appoximately 2-1/2 by 2 by 3/4 inches.

3.13 Identification marking. When specified (see 6.2), the bell and the buzzer shall be marked with the manufactures identification, part number, and National Stock Number.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be performed on each bell or buzzer when a first article sample is required (see 3.2 and 6.2). This inspection shall include the examination of 4.5 and the tests of 4.6. The first article may be a standard production item from the supplier's current inventory provided the bell or buzzer meets the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining bells or buzzers to be furnished under the contract.

4.4 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.5 and the test of 4.6.

4.4.1 Inspection lot. A lot shall consist of all bells and buzzers of the same type and style offered for delivery to the Government at one time under a specific contract.

4.4.2 Sampling for examination. A random sample of bells or buzzers shall be selected from each lot in accordance with inspection level II of MIL-STD-105. The Acceptable Quality Level (AQL) shall be 2.5 percent defective.

4.4.3 Sampling for tests. A random sample of bells or buzzers shall be selected from each lot in accordance with MIL-STD-105 at inspection level S-3. The AQL shall be 6.5 percent defective.

4.5 Examination. Each sample selected in accordance with 4.4.2 shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the supplier's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements.

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4.6 Test procedure. Each bell or buzzer composing the sample selected in accordance with 4.4.3 shall be tested. The testing shall include, but shall not necessarily be limited to, the test specified herein. The bell and the buzzer shall be operated for a period of 15 seconds duration at intervals of 30 seconds for 50 cycles of operation at the rated ac voltage, and for 50 cycles of operation at the rated dc voltage, without failure of any part thereof. During the operational test, the sound level shall be measureed with a sound-level meter (General Radio Type 1551-B or equal) to determine conformance to 3.6 or 3.7, as applicable.

4.7 Inspection of preparation for delivery. An inspection shall be made to determine that packaging, packing, and marking comply with the requirements in Section 5 of this specification. Defects shall be scored in accordance with table I. The sample unit shall be one shipping container packed as specified per 5.2, prepared for delivery and selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

TABLE I. Classification of preparation for delivery defects

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size location, sequence or method of application.
Materials	Any component missing or damaged. Packaging or packing material not as specified.
Workmanship	Inadequate application of components such as incomplete closure of container flaps. Distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. The bells shall be individually packaged in accordance with MIL-P-116, method IC-2. The buzzers shall be individually packaged in accordance with MIL-P-116, method IC-3.

5.1.1.1 Intermediate packaging. When specified (see 6.2), bells and buzzers of identical description shall be intermediate packaged in close-fitting boxes conforming to PPP-B-636, weather-resistant. Closure of the boxes shall be in accordance with appendix to PPP-B-636, method V.

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5.1.2 Level B - civil agencies. Each bell or buzzer shall be unit packaged in a close-fitting box conforming to PPP-B-566, or PPP-B-676.

5.1.3 Level C. The bells and buzzers shall be preserved and packaged in a manner that will insure adequate protection against deterioration and damage during shipment. This level may conform to the supplier's standard practice when such meets the requirements of this level.

5.2 Packing. The packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The bells and buzzers shall be packed in close fitting boxes conforming to PPP-B-601, overseas type; or PPP-B-636, V2s. The contents shall be blocked and braced to prevent movement within the box. The size or weight of the boxes shall not exceed the limitation of the applicable box specification. Closure and strapping shall be in accordance with the appendix of the applicable box specification.

5.2.2 Level B.

5.2.2.1 Military agencies. The bells and buzzers shall be packed as specified for level A, except the boxes shall conform to PPP-B-601, domestic type; or PPP-B-636, V3c or V3s. When specified (see 6.2), fiberboard boxes shall conform to PPP-B-636, class domestic.

5.2.2.2 Civil agencies. The bells and buzzers shall be packed in a close-fitting fiberboard box conforming to PPP-B-636, class domestic, in quantities as specified (see 6.2). Closure of the boxes shall be in accordance with the appendix to the box specification.

5.2.3 Level C. The bells and buzzers shall be packed in a manner which will insure arrival at destination in a satisfactory condition. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, unit and intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The bell and the buzzer covered by this specification is intended for use as signal devices in interior communication circuits with a standard bell ringing transformer or dry cell batteries at the voltages specified in 3.6.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type and style required (see 1.2.1).
(c) When first article is required for inspection and approval (see 3.2, 4.3, and 6.5).
(d) Voltage, if other than specified (see 3.6).
(e) Identification marking, if required (see 3.13).
(f) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).
(g) When intermediate packaging is required (see 5.1.1.1).
(h) When domestic fiberboard boxes are to be used (see 5.2.2.1).
(i) Quantities required in the shipping container for civil agencies shipments (see 5.2.2.2).

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should definitely be stated, the specification to apply in all other respects.

6.4 Federal specifications do not include all types and styles of the commodity indicated by the title of the specification, or which are commerically available, but are intended to cover the kinds which are suitable for Federal Government requirements.

6.5 First article - military agencies. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the Armed Services Procurement Regulations. The first article may be a standard production item from the supplier's current inventory as specified in 4.2.1. The first article should consist of one complete bell or buzzer of each type and style included under a specific contract. The contracting officer should include specific instructions in all procurement instruments, regarding arrangments for examinations, tests, and approval of the first article.

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MILITARY INTEREST:

Custodian:

Navy - YD

User activity:

Army - EL

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
DC GOVT - DCG
DOT - FIS
INTERIOR - BPA

Preparing activity:

Navy - YD

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Orders for this publication are to be placed with General Services Administration, acting as agent for the Superintendent of Documents. See section 2 of this specification to obtain copies of documents referenced herein. Price 70 cents each.

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